

Loops in Python

There are two loops: 1. while loop 2. for loop

While Loops

used when you don't know how many times to repeat: you loop until a condition is met. ex. retrying a task until it succeeds, waiting for user input

```
In [4]: print('Data Science')
        print('Data Science')
        print('Data Science')
        print('Data Science')
        print('Data Science')

# Here we had to print the statement for 5 times only
# if have to print it for hundreds or thousands time it will be difficult and time
# But this can be done esily using a while loop
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [7]: i = 1                                # initializing          # giving value to i
        while i <= 5:                        # till i <= 5 loop follows
            print('Data Science')            # indentation (4 space) # print given statement
            i += 1                            # for evey repeat of loop increment value by on
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [11]: i = 1                               # initializing          # giving value to i
         while i <= 5:                       # till i <= 5 loop follows
             print('Data Science', i)        # print given statement
             i += 1                          # for evey repeat of loop increment value by on
         # by printing i alongside a statement we can track the incremented value in each re
```

Data Science 1
Data Science 2
Data Science 3
Data Science 4
Data Science 5

```
In [9]: i=5                                # initializing
        while i>=1:                          # condition
            print('Data Science')
            i -= 1                            # decrement by 1
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [13]: i=5                # initializing
         while i>=1:        # condition
             print('Data Science', i)
             i -= 1         # decrement by 1

         # by printing i alongside a statement we can track the decremented value in each re
```

Data Science 5
Data Science 4
Data Science 3
Data Science 2
Data Science 1

Nested While

```
In [16]: i = 1
         while i<=5:
             print('Data Science')
             j = 1
             while j <= 4:                # nested while (while inside another while)
                 print('technology')
                 j +=1
             i += 1
```

Data Science
technology
technology
technology
technology
Data Science
technology
technology
technology
technology
Data Science
technology
technology
technology
technology
Data Science
technology
technology
technology
technology
Data Science
technology
technology
technology
technology

the output we got is too lengthy ,how to make it print in one line let's see

```
In [35]: i = 1
while i<=5:
    print('Data Science', end = " ") # when we mention end new line will not get cr
    j = 1
    while j <= 4:                    # nested while
        print('technology', end= " ")
        j +=1
    i += 1
    print()
```

*# when we write the last line code print() then only the printed text comes in orgn
if do not write it or wrte outside while loop then it will not be in organised wa*

```
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
```

```
In [37]: # Let's use while loop using some numbers
```

```
i = 1
while i <= 4:
    j = 0
    while j <= 3:
        print(i*j, end= " ")
        j += 1
    i += 1
    print()
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

for loop

used when you know how many times to repeat ex. looping over a list of names, processing number of files, counting from 1 to 100 break: break/stop the loop if condition met continue: skip the condition satisfying number and continue further pass: Do nothing here, but this block is intentional.

```
In [45]: name = 'nit'

for i in name:
    print(i)
```

```
n
i
t
```

```
In [47]: name1 = (1, 3.5, 'hello')    # i want print the values individually
```

```
for i in name1:  
    print(i)
```

1
3.5
hello

```
In [49]: for i in range(5):           # range(5) = 0,1,2,3,4  
        print(i)
```

0
1
2
3
4

```
In [60]: for i in range( 1,10,3):     # numbers from 1 to 9 with step count of 3  
        print(i)  
  
        # 1,(1+3),(1+3+3)
```

1
4
7

```
In [68]: for i in range(1,51):  
        if i%5 == 0:  
            print(i)
```

5
10
15
20
25
30
35
40
45
50

```
In [70]: for i in range(1,51):  
        if i%5 != 0:  
            print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

```
In [72]: for i in range (1,11):  
         print(i)
```

1
2
3
4
5
6
7
8
9
10

```
In [81]: for i in range (1,11):  
        if i == 5:  
            break                # break/stop the loop if condition met  
        print(i)
```

1
2
3
4

```
In [83]: for i in range (1,11):  
        if i == 7:  
            break  
        print(i)
```

1
2
3
4
5
6

```
In [85]: for i in range (1,11):  
        if i == 6:  
            continue            # skip the condition satisfying number and continue f  
        print(i)
```

1
2
3
4
5
7
8
9
10

```
In [89]: for i in range (1,11):  
        if i == 5:  
            continue            # skip the condition satisfying number and continue f  
        print("hello",i)
```

hello 1
hello 2
hello 3
hello 4
hello 6
hello 7
hello 8
hello 9
hello 10

```
In [93]: for i in range(1,11):  
        pass
```

```
In [99]: for i in range(1,51):  
        if i%3 == 0:
```

```
print(i)  
print("end")
```

3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end

In [102...

```
for i in range(1,51):  
    if i%3 == 0:  
        continue      # if i%3 == 0 skip the number  
    print(i)  
print("end")
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

In [110...

```
for i in range(1,51):
    if i%3 == 0 and i%5 == 0:          # and- both condition should satisfied # summe
        continue                     # if i%3 == 0 and i%5 == 0 :skip the number
    print(i)
print("end")

# it will skip all value which is divisible by 15 ( divided by 3 and 5 both) (3*5)
```


1
2
3
4
5
6
7
8
9
10
11
12
13
14
16
17
18
19
20
21
22
23
24
25
26
27
28
29
31
32
33
34
35
36
37
38
39
40
41
42
43
44
46
47
48
49
50
end

In [114...

```
for i in range(1,51):  
    if i%3 == 0 or i%5 == 0:          # or- any one of given condition should satisfi  
        continue                    # if i%3 == 0 or i%5 == 0 :skip the number  
    print(i)  
print("end")  
  
# it will skip all value which is divisible by 3 or 5
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end

In [116... *# i dont want to print the values which are odd numbers that means print only even*

```
for i in range(0,20):  
    if i%2 == 0:           # after divided by 2, reminder=0 : even number  
        print(i)
```

0
2
4
6
8
10
12
14
16
18

In [126... *# print only odd numbers*

```
for i in range(0,20):  
    if i%2 == 0:           # after divided by 2, reminder=0 : even number  
        pass              # pass: pass/do nothing if i%2==0  
    else:  
        print(i)
```

1
3
5
7
9
11
13
15
17
19

Nested for loop

In [130...

```
print("# # # #")
print("# # # #")
print("# # # #")
print("# # # #")
```

```
# # # #
# # # #
# # # #
# # # #
```

In [134...

```
for j in range(4):
    print("#",end=" ")
```

```
# # # #
```

In [140...

```
for j in range(4):
    print("#",end=" ")
    print()
```

```
for j in range(4):
    print("#",end=" ")
    print()
```

```
for j in range(4):
    print("#",end=" ")
    print()
```

```
for j in range(4):
    print("#",end=" ")
    print()
```

need to write multiple time instead of this can use nested for loop

```
# # # #
# # # #
# # # #
# # # #
```

In [142...

```
for i in range(4):
    for j in range(4):
        print("#",end=" ")
    print()
```

nested for

use debug code in pycharm: pycharm> paste code> right click>debug test1> f8 for e

```
# # # #  
# # # #  
# # # #  
# # # #
```